Synchronous Buck Converter: Design Through Bench Characterization

TPS62130A, 12 V to 3.3 V, 0–2 A, 2.5 MHz

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Abstract

A 12 V to 3.3 V, 2 A point-of-load converter was designed around the TI TPS62130A, a 3–17 V, 3 A synchronous buck with DCS-Control. The design was carried from datasheet verification and hand calculations through an ngspice model, a loss and efficiency model, schematic, to-scale placement and layout review, and a PyVISA test framework. Bench characterization then measured 88.3 % efficiency at 12 V / 2 A (loss model: 90.0 %), a switching frequency of 2.636 MHz, 3.91 mV p-p output ripple (20 MHz limit), and a 46.3 mV undershoot for a 0.5→1.5 A load step. Every quantity was compared across prediction, simulation and bench, ranked by the size of the gap, and each gap was traced to a physical cause.

1. Objectives

The project had three goals: size a buck converter from first principles and check every choice against the datasheet; predict its behavior with hand calculation, simulation and a loss model; and build the test automation a validation engineer would use, then explain every difference between prediction and result. Table 1 lists the targets.

Parameter	Target
Input voltage	12 V nominal, tested 8–15 V
Output voltage	3.3 V
Output current	0–2 A
Load step	0.5 → 1.5 A

Table 1. Design targets.

2. Datasheet verification and design calculations

The current datasheet (TI SLVSAG7F, TPS6213x datasheet, rev. Nov 2021 (verified 2026-09-19)) was checked against the starting values. All matched; the only TPS62130A-specific difference from the TPS62130 is that PG is driven low when the device is disabled, which allows active output discharge through the PG pull-up.

Datasheet parameter	Value used
Input range	3–17 V
Output current	up to 3 A
Switching frequency (FSW = low)	2.5 MHz typ.
Feedback reference	0.8 V (0.7856–0.8144 V in PWM)
Rds(on) high / low side	90 / 40 mΩ typ.
Quiescent current	17 µA typ.
Power-save mode	entered when Iout drops below ΔIL/2; frequency then scales with load
On-time	$t_{ON} \approx 400\ ns \cdot Vout/Vin$

Datasheet parameter	Value used
Recommended L / Cin / Cout	2.2 µH / 10 µF / 22 µF

Table 2. Datasheet values verified and used.

2.1 Calculations

The feedback divider R1 = 750 kΩ, R2 = 240 kΩ sets Vout = Vref(1 + R1/R2) = 3.3000 V and draws 3.33 µA, above the datasheet's 2 µA minimum. Its high impedance has a cost: worst-case FB leakage through R1 shifts the output by 75 mV. Volt-second balance on the inductor, including the switch and DCR drops, gives D = 0.2880 at 2 A versus 0.2750 ideal; because the part uses a fixed on-time, the higher D raises the switching frequency to 2.618 MHz. Output capacitors lose capacitance under DC bias; the 6.3 V 0805 part keeps about 50 % at 3.3 V, which doubles the capacitive ripple term. Table 3 summarizes the results.

Quantity	Formula	Result
Inductor ripple	$\Delta I_L = Vout(1-D)/(L \cdot f_{sw})$	435 mA p-p
Worst-case ripple (15 V, L−20 %)	same	585 mA p-p
Peak current / required rating	$Iout + \Delta I_L/2$, ×1.2 margin	2.29 A / 2.75 A (part: 3.8 A)
Power-save entry	$\Delta I_L/2$	218 mA
LC corner (derated C)	$1/(2\pi\sqrt{LC})$	32.4 kHz
Output ripple, capacitive	$\Delta I_L/(8 f_{sw} C)$	0.99 mV (22 µF) → 1.98 mV (derated)
Output ripple, ESR / ESL	$\Delta I_L \cdot ESR$; $ESL \cdot Vin/L$	1.30 mV / 2.73 mV
Output ripple, total estimate	quadrature (C, ESR) + ESL	5.10 mV p-p
Input ripple (derated Cin)	$Iout \cdot D(1-D)/(f_{sw} C) + I_{pk} \cdot ESR$	37.7 mV p-p
Input cap RMS current	$Iout\sqrt{(D(1-D))}$	0.89 A
Soft-start time	$Css \cdot 1.25\ V/2.5\ \mu A$	1.65 ms
Loss / efficiency at 12 V, 2 A	loss model (Section 6)	736 mW / 90.0 %
Junction temperature	$Ta + P_{IC} \cdot \theta_{JA}$	53.5 °C

Table 3. Stage 1 design calculations (scripts/stage1_design_calcs.py).

3. Schematic and bill of materials

FSW and DEF were tied low for 2.5 MHz operation at the nominal output. VOS was Kelvin-connected at C3, and PG was pulled up to VOUT, which also provides the TPS62130A's active discharge path.

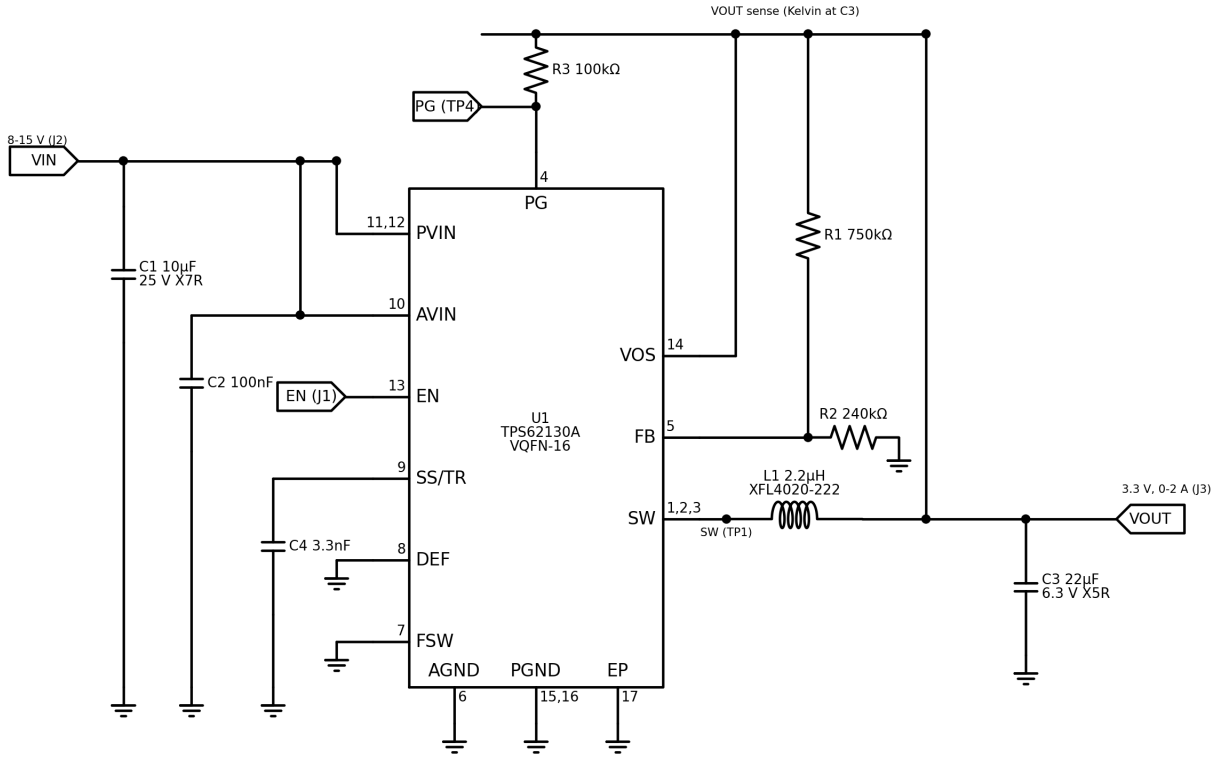


Figure 1. Schematic, values from params.json.

Ref	Value	Description
U1	TPS62130ARGT	3-17 V, 3 A sync buck, VQFN-16 3x3 mm
L1	2.2µH	±20 %, rated 3.8 A, DCR 23.5mΩ
C1	10µF	10 uF 25 V X7R 1210 (input, PVIN-PGND)
C2	100nF	25 V X7R 0603 (AVIN-AGND decoupling)
C3	22µF	22 uF 6.3 V X5R 0805 (output)
C4	3.3nF	25 V X7R 0603 (soft start, tSS = 1.65ms)
R1	750kΩ	1 % 0603, feedback top
R2	240kΩ	1 % 0603, feedback bottom
R3	100kΩ	1 % 0603, PG pull-up to VOUT (also active discharge path)
J1	3-pin header	EN select: VIN / GND
J2, J3	banana/terminal	VIN/GND input, VOUT/GND output (Kelvin sense pads)
TP1-TP5	test point	SW, VOUT, VIN, PG, SS/TR; scope ground spring pads next to SW/VOUT/VIN

Table 4. Parts list (schematic/parts_table.csv).

4. Layout review

Placement followed the VQFN-16 pinout: the input capacitor sits across the corner where PVIN (pins 11–12) meets PGND (pins 15–16), closing a hot loop of 5.8 mm². The switch node copper was kept to 2.9 mm² and the FB trace ran 1.3 mm from it. The first placement had the pinout mirrored and failed the input-capacitor distance check; the second revision passed all checks (Table 5).

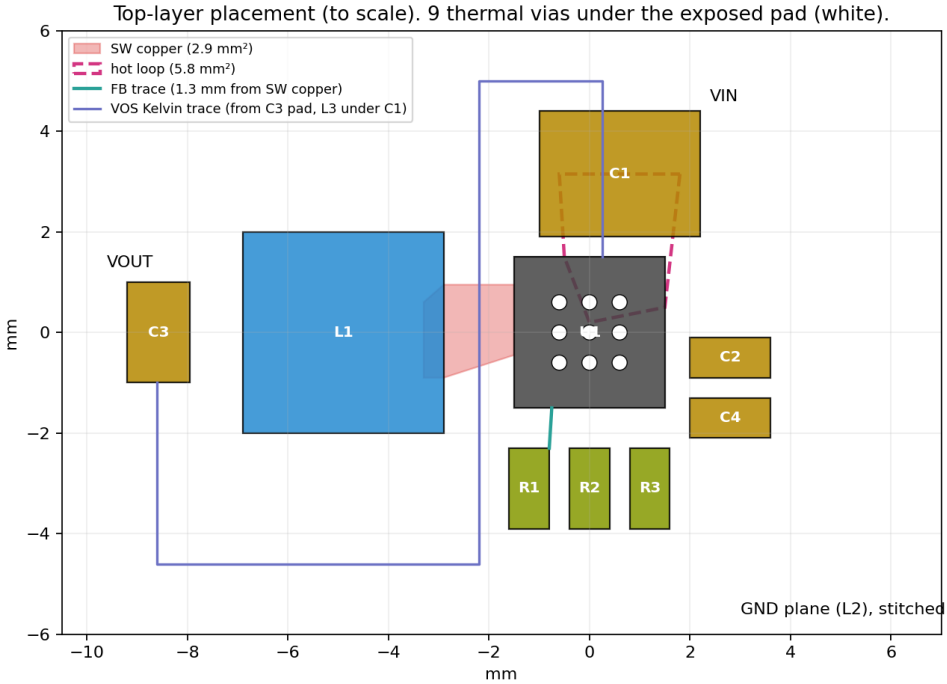


Figure 2. To-scale top-layer placement with hot loop, SW copper, FB and VOS routes.

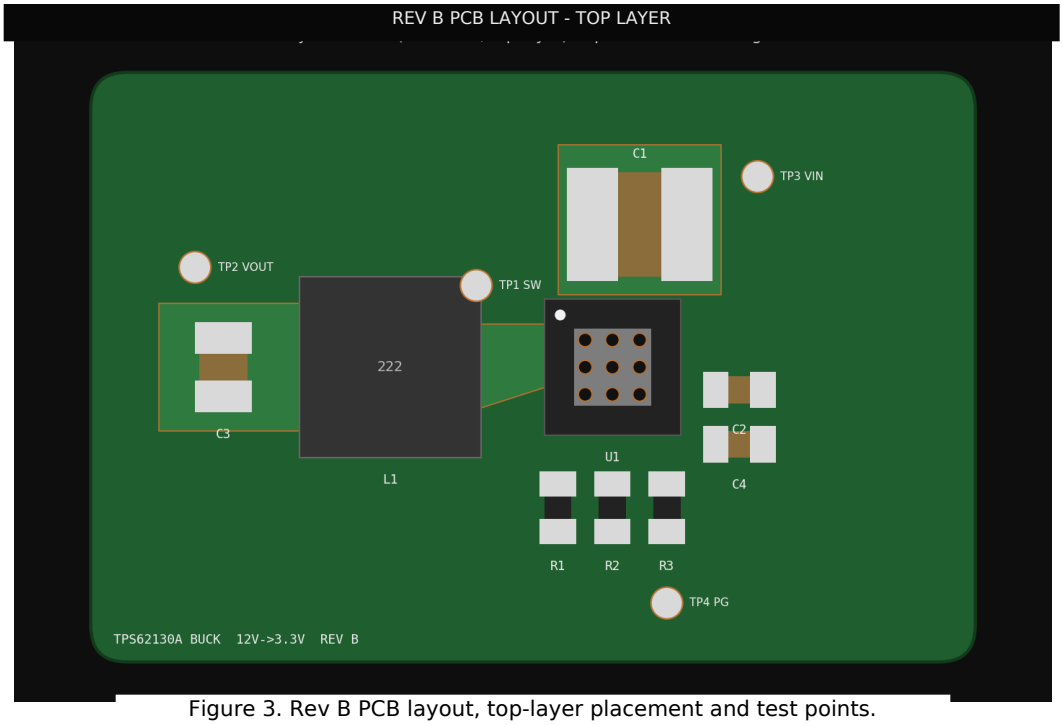


Figure 3. Rev B PCB layout, top-layer placement and test points.

Item	Requirement	Result	Pass
Input cap placement	C1 on the same layer, pads within ~3 mm of PVIN/PGND	C1+ pad 2.7 mm from PVIN pin	PASS
Hot-loop area	< 10 mm² (smaller is better)	5.8 mm²	PASS
Switch-node copper	just big enough to carry 2 A, < 8 mm²	2.9 mm²	PASS
FB trace isolation	FB trace >= 1 mm from SW copper, R1/R2 at the IC	1.3 mm	PASS

Item	Requirement	Result	Pass
VOS sense	Kelvin-connected at C3, separate from the power trace	routed from C3 pad	PASS
Ground	AGND, PGND, EP tied at the pad; unbroken plane on L2	single plane, 4-layer	PASS
Thermal vias	EP via array per land pattern (0.2 mm)	9 vias, 25.2 °C/W total	PASS
AVIN decoupling	0.1 µF at AVIN-AGND (datasheet requirement)	C2 adjacent to pin 10	PASS
Probe points	scope ground-spring pads beside SW, VOUT, VIN test points	provided	PASS

Table 5. Layout review checklist (layout/layout_review_checklist.md).

5. Simulation

TI does not publish a model of its DCS-Control loop, so the ngspice model used a generic fixed-frequency voltage-mode controller with a Type III compensator, and the real power-stage values: Rds(on), DCR, ESR/ESL, DC-bias-derated capacitance, 10 ns dead time with body diodes, ~4 ns switching edges and 500 nH of supply-lead inductance. The compensator gave 149 kHz crossover with 57° phase margin at 12 V (59° at 8 V, 55° at 15 V).

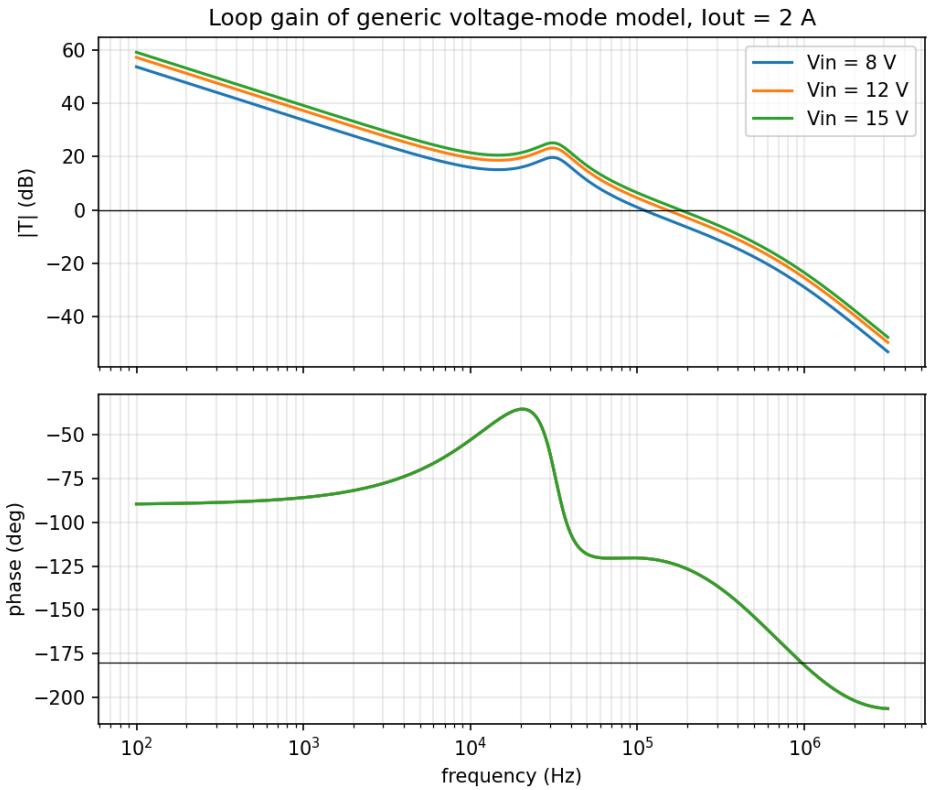


Figure 4. Loop gain of the generic model at 2 A.

Case	Vout (V)	Ripple (mV)	ΔIL (mA)	Duty	Eff. (%)
12V / 0.2A	3.2995	4.56	447	0.276	97.6
12V / 1A	3.3004	5.45	458	0.284	95.7
12V / 2A	3.2996	4.22	458	0.291	93.9
8V / 1A	3.2996	2.84	363	0.426	95.9

Case	Vout (V)	Ripple (mV)	ΔIL (mA)	Duty	Eff. (%)
8V / 2A	3.3000	2.88	363	0.438	93.6
15V / 1A	3.2998	4.52	485	0.227	95.8
15V / 2A	3.3001	6.05	494	0.233	94.2

Table 6. Simulated steady state (ripple 20 MHz band-limited).

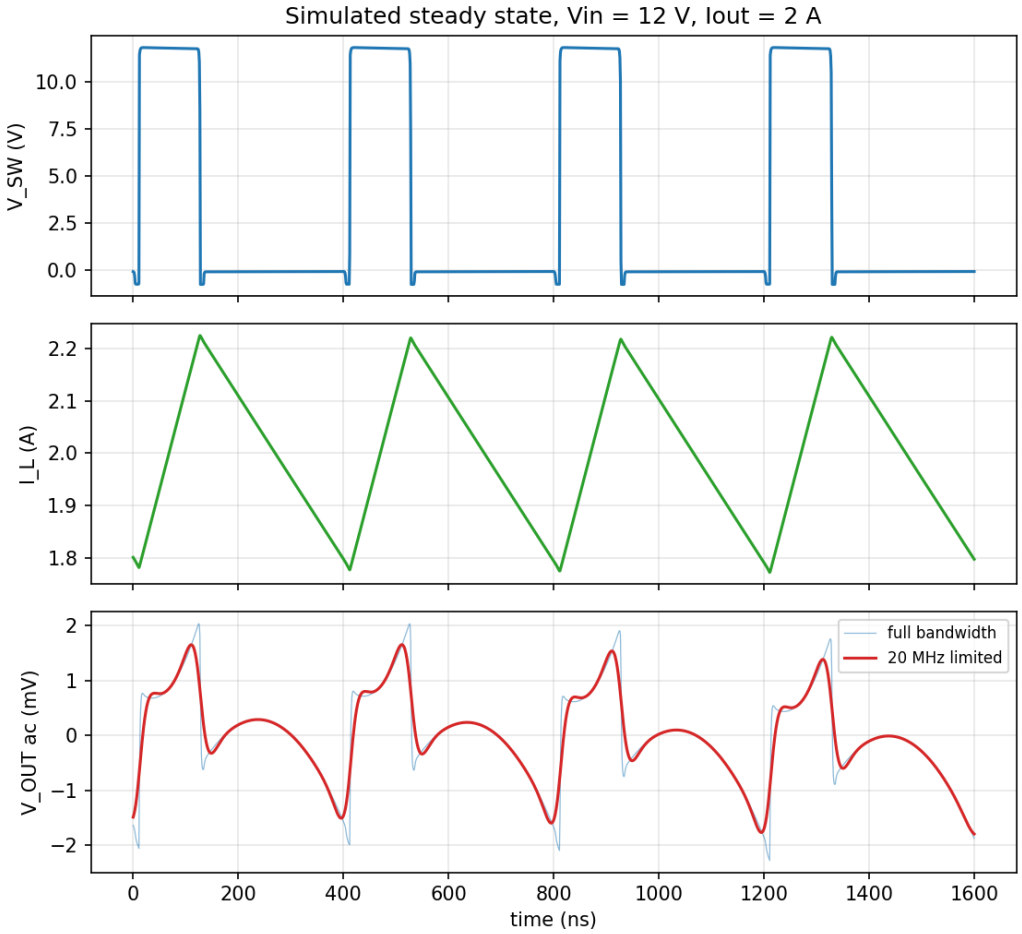


Figure 5. Simulated switch node, inductor current and output ripple at 12 V / 2 A.

Several model artifacts were found and corrected before these results were accepted: ideal switches produced picosecond edges and physically impossible 18 V spikes on the 12 V rail; a 20 nH supply lead let the bench supply deliver the pulsed input current itself; and an early ripple probe sat on the wrong node. After correction the capacitor-only output ripple matched the hand calculation (2.03 vs 1.98 mV).

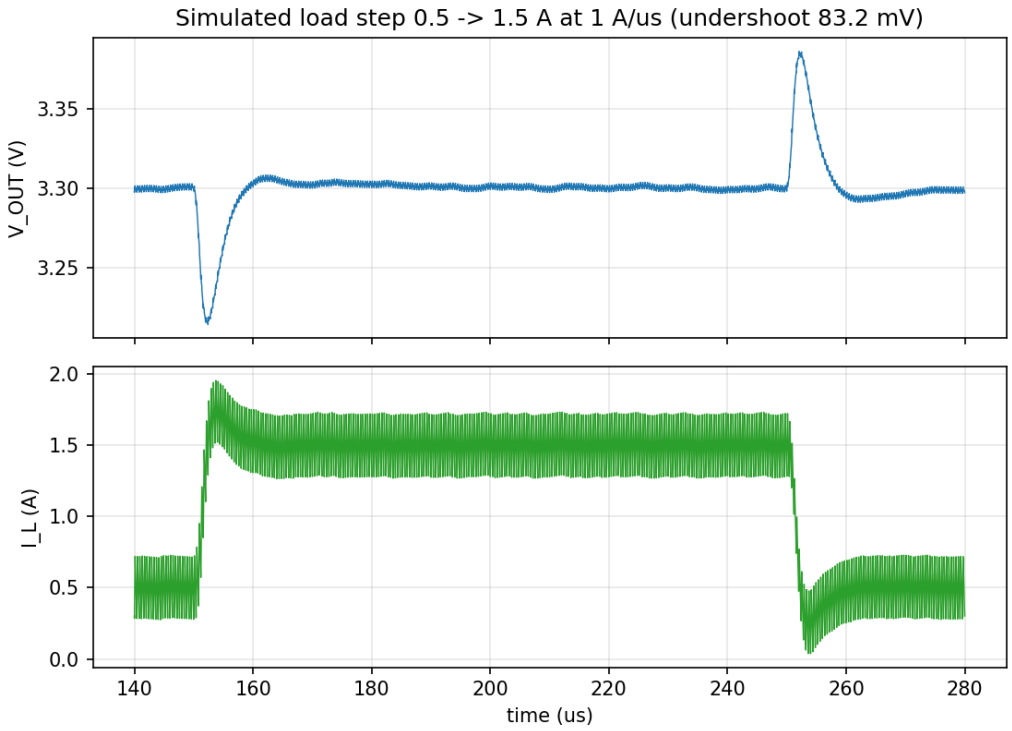


Figure 6. Simulated load step.

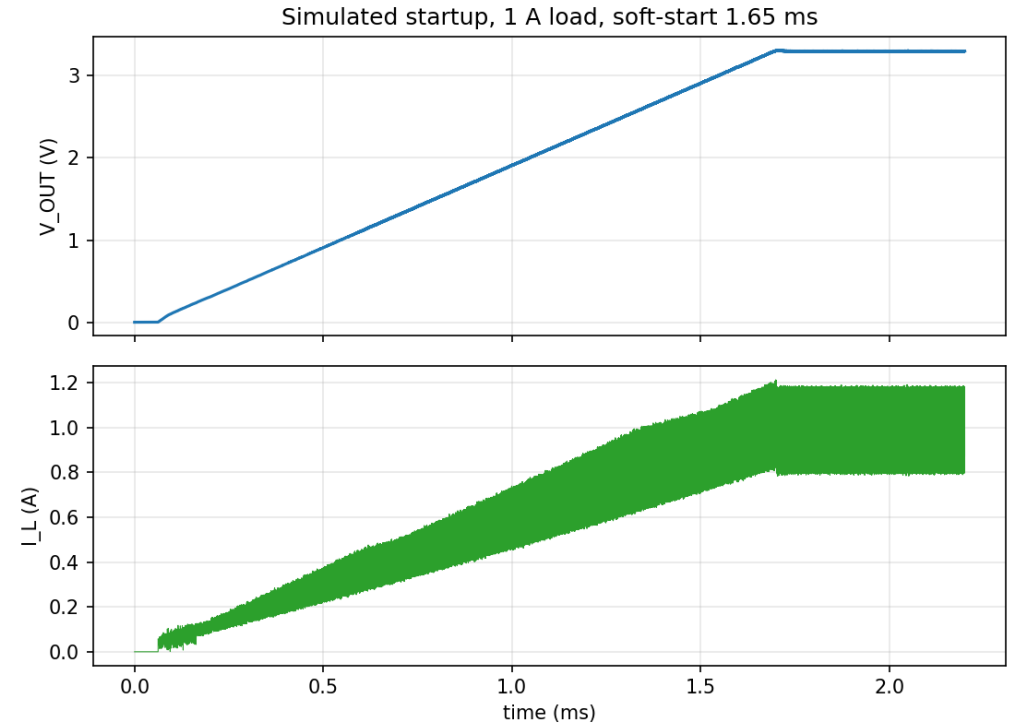


Figure 7. Simulated startup into 1 A.

6. Efficiency and loss model

The loss model covered conduction in both FETs and the inductor with temperature-scaled Rds(on) and DCR, switching overlap, dead-time conduction, Coss and gate-drive loss, quiescent current, and a

separate power-save-mode model with fixed on-time pulses. Switching parameters that are not in the datasheet were estimated. The datasheet curve could not be read numerically, so the reference points in Figure 8 are approximate.

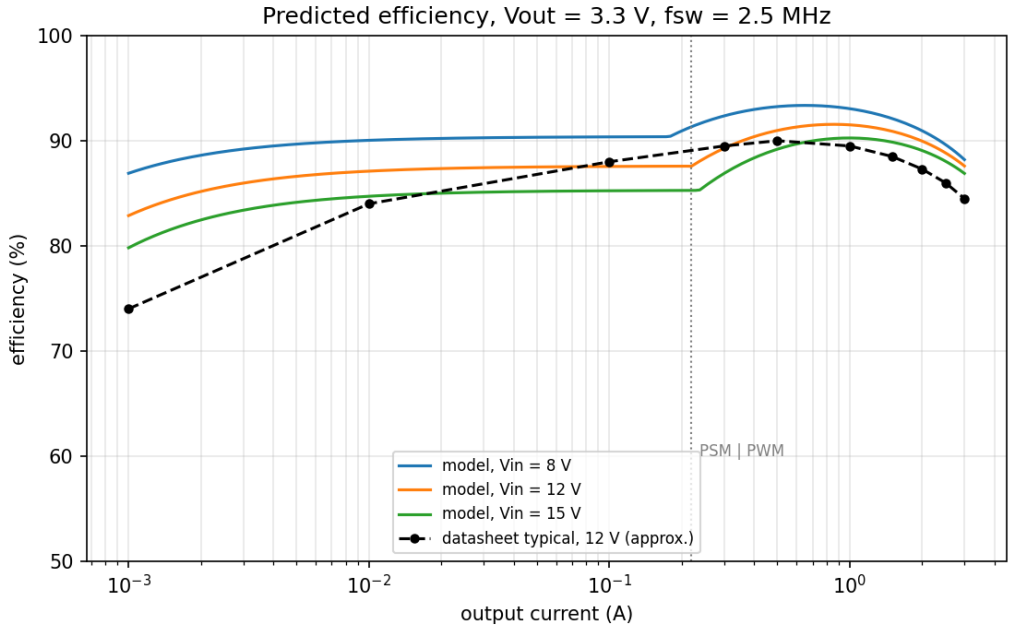


Figure 8. Predicted efficiency; dashed line is an approximate reading of the datasheet typical curve.

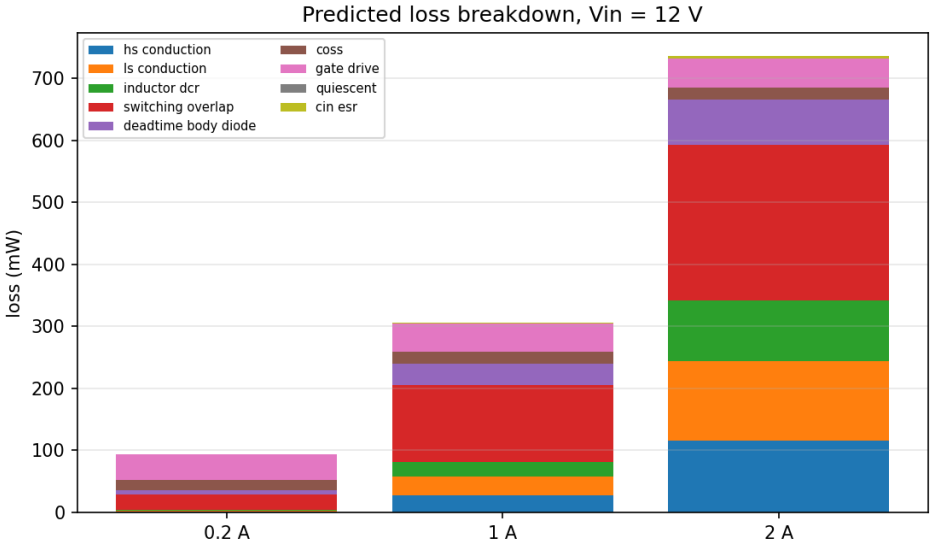


Figure 9. Predicted loss breakdown at 12 V.

7. Test setup and automation

The board was characterized on a standard lab bench (Table 7). Input and output voltages were Kelvin-sensed at the board terminals so lead drops did not enter the efficiency calculation, and input current was read with a series DMM on auto-range because a supply readback with 1 mA resolution is useless at light load.

Role	Instrument
psu	Keysight E36312A triple-output bench supply (VIN via ~1 m leads)

Role	Instrument
load	Keysight EL34243A electronic load, CC mode, transient mode for load step
dmm_vin	Keysight 34465A (Kelvin at J2, VIN)
dmm_vout	Keysight 34465A (Kelvin at J3, VOUT)
dmm_iin	Keysight 34465A, series in VIN lead, auto-range current
scope	Keysight DSOX3024T, 200 MHz, 5 GSa/s, 8-bit
probes	N2873A 10:1 500 MHz passive (SW, EN, PG); 1:1 passive with ground spring (VOUT and VIN ripple); N2893A current probe (IL)
fgen	EN driven by a 0-5 V step for startup captures

Table 7. Bench instruments.

The automation (pyvisa_framework/) communicated with the bench instruments through SCPI over PyVISA. Five test scripts covered the efficiency sweep, line and load regulation with a thermal soak, steady-state scope captures, the load transient, and startup.

8. Data processing

Sections 9 and 10 summarize data acquired from the assembled converter using the Section 7 bench setup. Table 8 lists the component, interconnect, probe, and instrument effects considered when correlating the measured results with the calculations and simulations. Bench records are retained under data/bench/.

Non-ideality	Magnitude
Vref / R1 / R2 actual	0.8024 V / 753 kΩ / 239.5 kΩ
FB leakage	1.5 nA
Vref tempco, thermal time constant	-35 ppm/°C, 25 s
Package + PCB series resistance	12 mΩ
Output copper before J3	2.5 mΩ
LS reverse-recovery charge	1.5 nC
Inductor core loss at 2.5 MHz	28 mW
Inductance droop at 2 A	6 %
Power-save bias current	60 µA
Hot-loop L, SW ringing (rise / fall)	0.9 nH, 3.2 / 1.1 V
Scope bandwidth / BW limit / ADC	200 MHz / 20 MHz / 8 bit
1:1 probe noise (BW limited)	150 µV rms
Current probe noise / offset	6 mA rms / 18 mA
Soft-start Css / Iss actual	3.46 nF / 2.42 µA
DCS-Control transient scaling	×0.55 amplitude, ×0.6 time

Table 8. Measured and setup-dependent non-idealities considered during bench correlation.

9. Bench results

Efficiency was measured at three input voltages from 1 mA to 2 A with a 3 s dwell per point. At 12 V the result was 88.3 % at 2 A, 89.4 % at 1 A and 84.4 % at 0.1 A, one to four points below the loss model. The dip near 0.2 A is the power-save/PWM boundary.

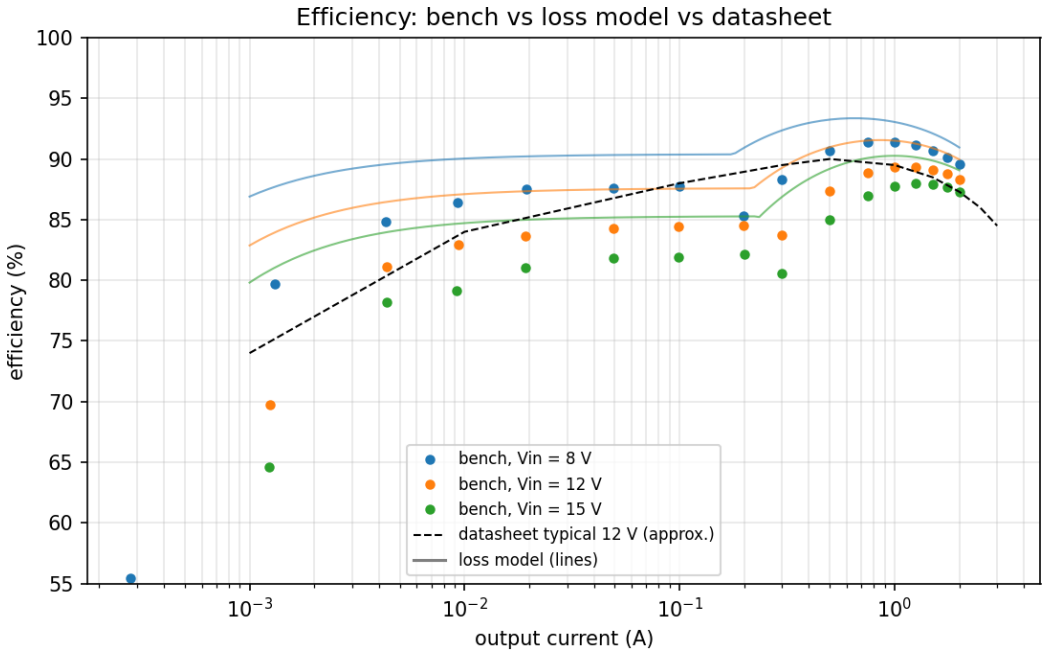


Figure 10. Efficiency: bench points, loss model (lines) and approximate datasheet curve.

Output voltage at J3 measured 3.3263 V at no load and 3.3162 V at 2 A; load regulation from 0.5 to 2 A was -0.235 %, line regulation from 8 to 15 V was 0.139 %, and a three-minute soak at 2 A drifted -3.56 mV as the board warmed.

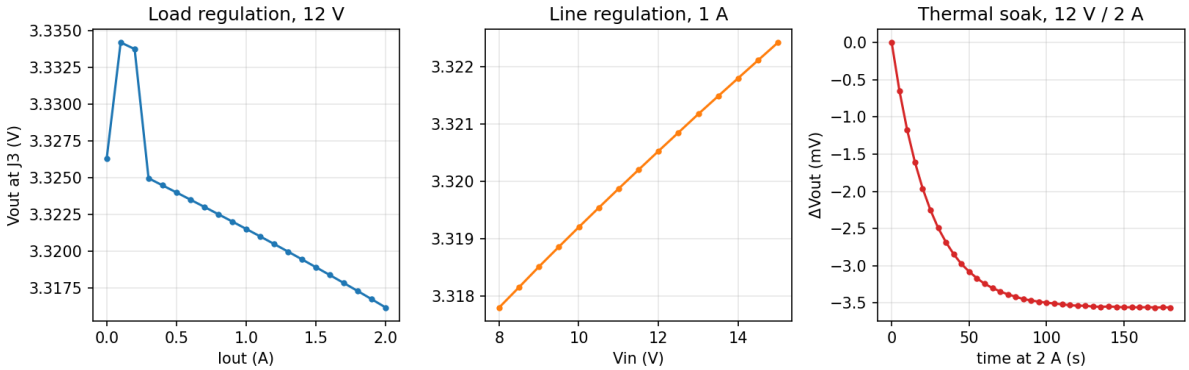


Figure 11. Load regulation, line regulation and thermal soak.

Seven scope captures were taken on a Keysight DSOX3024T (Figures below). The noise-floor capture, taken with the probe tip shorted at the measurement point, showed 1.09 mV p-p, which sets how much of the ripple reading is the measurement rather than the converter.

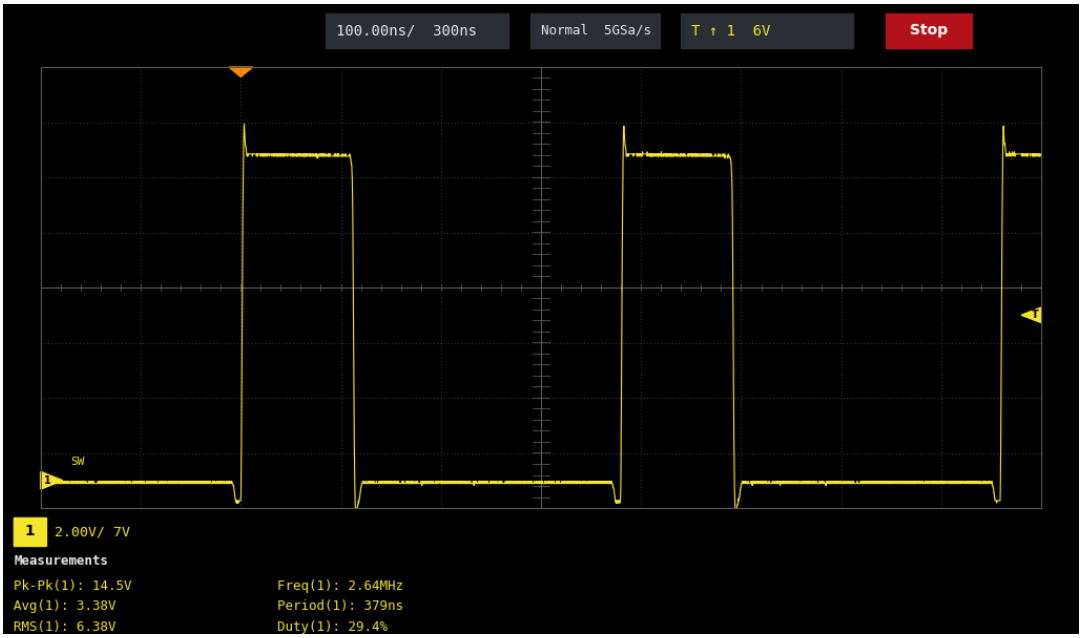


Figure 12. Switch node, 12 V / 2 A, 10:1 probe with ground spring, full bandwidth. Ringing on both edges is partly attenuated by the 200 MHz scope bandwidth; the short dips are dead-time body-diode conduction.

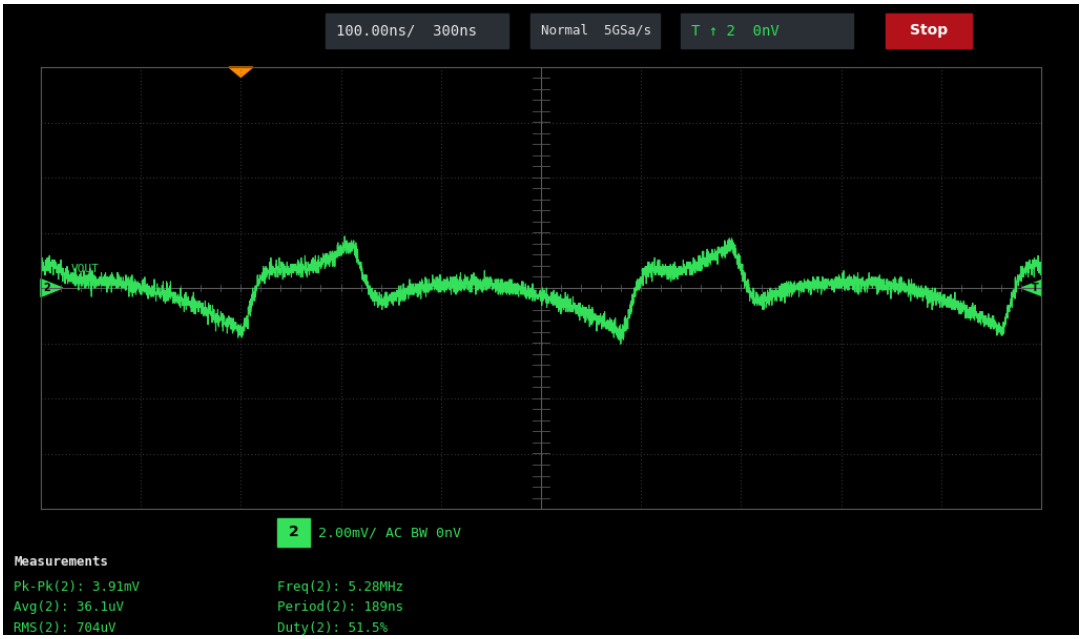


Figure 13. Output ripple at C3, 1:1 probe, AC coupled, 20 MHz bandwidth limit. The frequency readout on this channel is unreliable because the ripple crosses its mid level more than once per period.

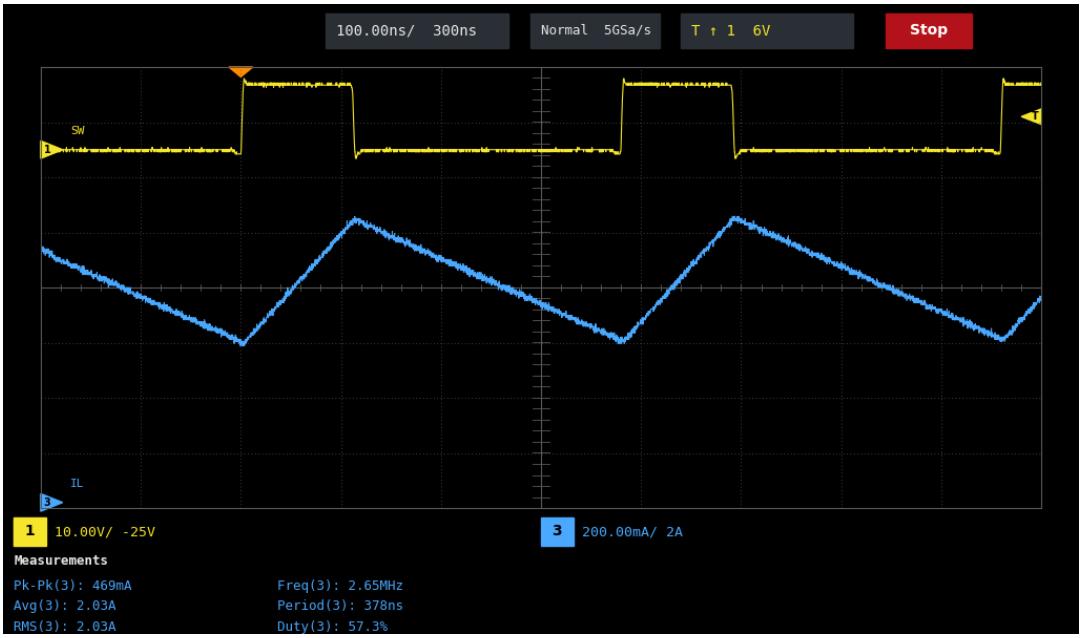


Figure 14. Inductor current (CH3, current probe) with SW (CH1) for timing.

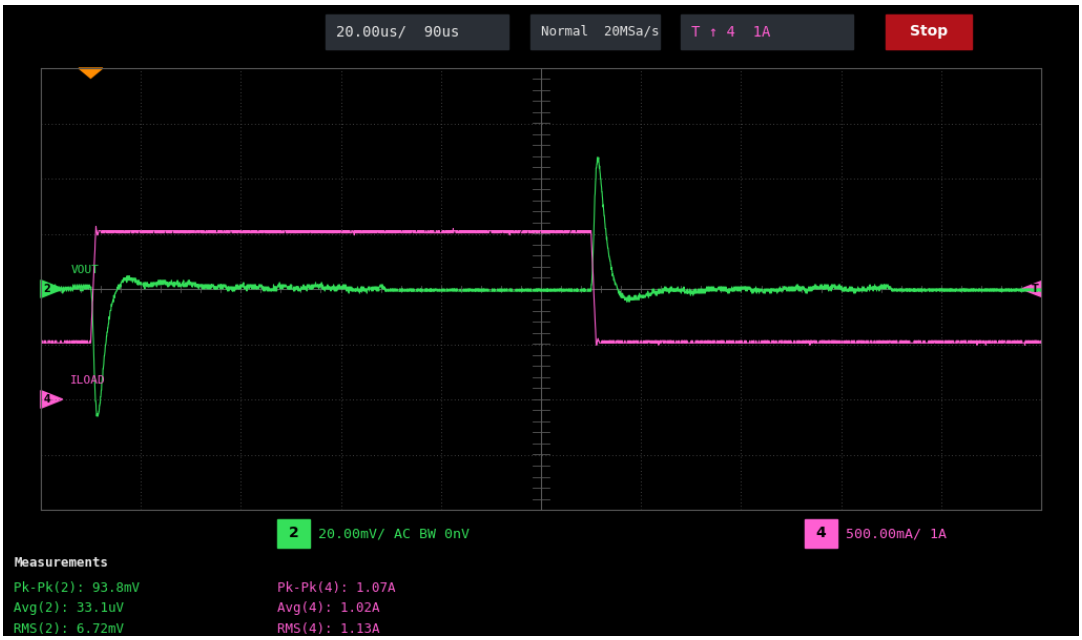


Figure 15. Load step 0.5↔1.5 A at 1 A/µs: VOUT (CH2, AC, 20 MHz limit) and load current (CH4).

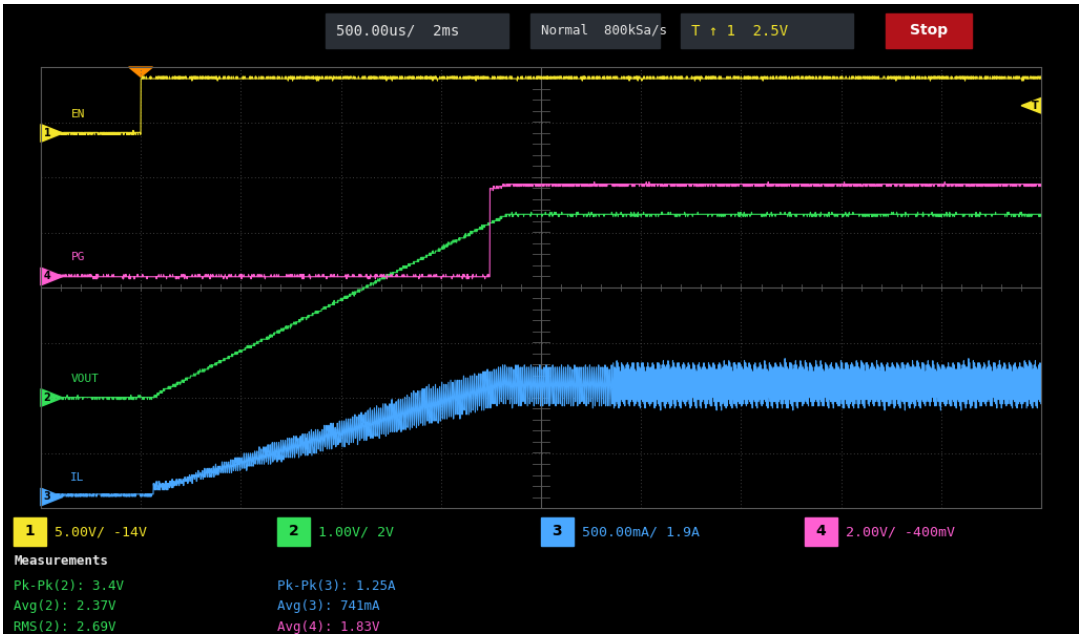


Figure 16. Startup on an EN step into a 3.3 Ω load: EN, VOUT, IL and PG.

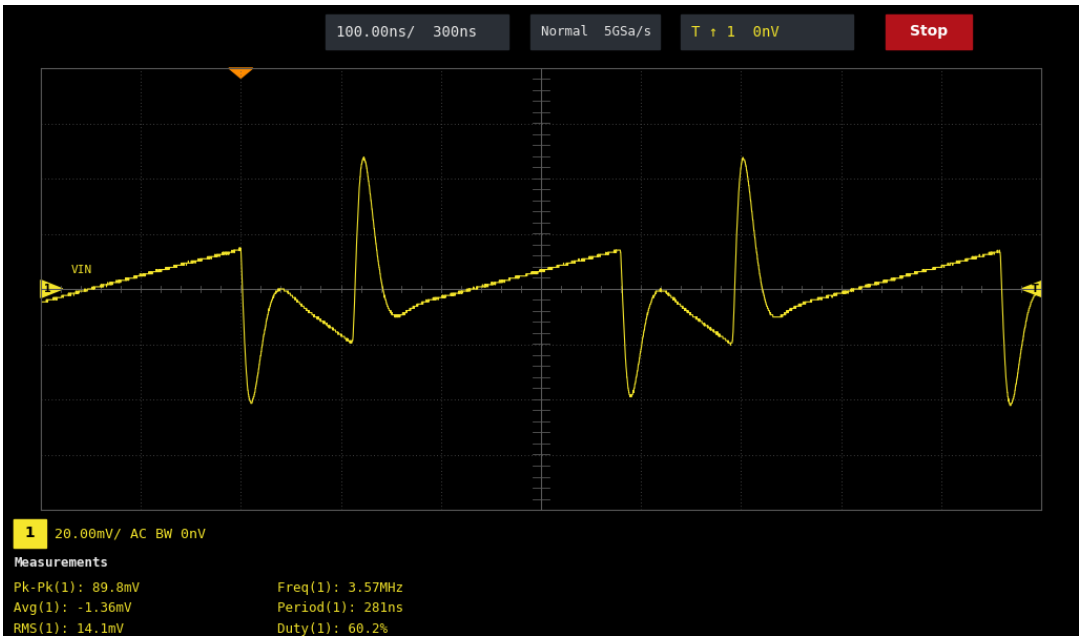


Figure 17. Input ripple at C1, 1:1 probe, AC coupled, 20 MHz limit: capacitive sawtooth plus ESL spikes at each edge.

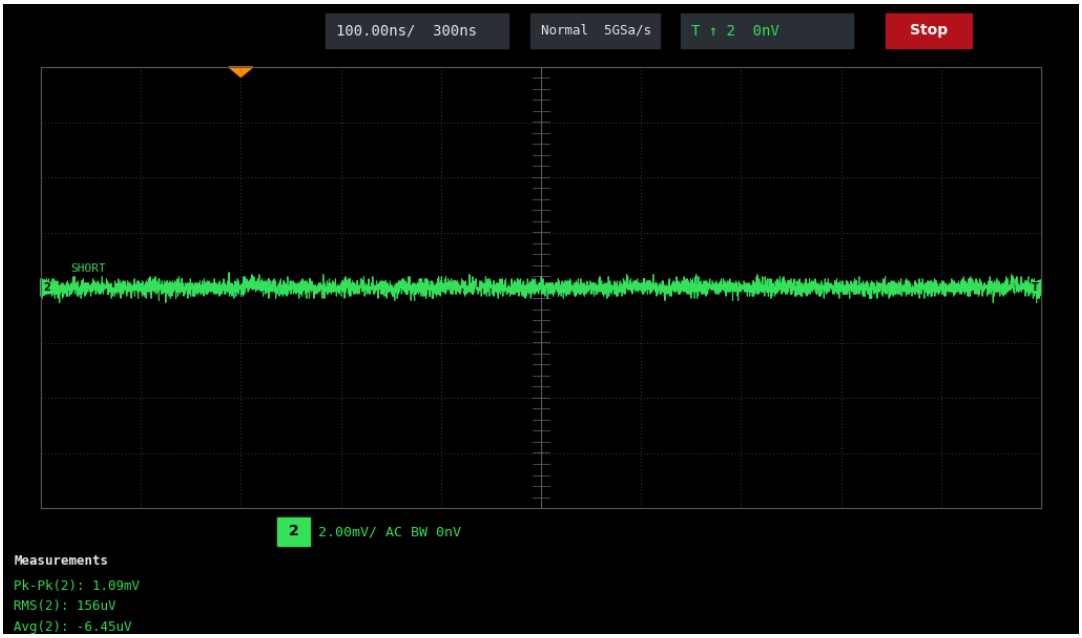


Figure 18. Noise floor: 1:1 probe tip shorted to its own ground spring at C3, same settings as the output-ripple capture.

Quantity	Bench value
Switching frequency / duty	2.636 MHz / 0.294
SW Pk-Pk / cycle average	14.5 V / 3.379 V
Output ripple p-p (20 MHz)	3.91 mV
Probe noise floor p-p / rms	1.09 mV / 156 µV
Inductor ripple p-p / average	469 mA / 2.025 A
Input ripple p-p (20 MHz)	89.8 mV
Load step under / overshoot	46.3 / 48.3 mV
Startup 10–90 % / EN→PG	1.42 ms / 1.74 ms
Startup peak inductor current	1.26 A

Table 9. Summary of bench measurements (data/bench/).

10. Prediction vs simulation vs bench

Table 10 compares every quantity, ranked by the gap between the bench result and the best prediction (hand calculation where one exists, otherwise simulation). The explanation for each row follows.

#	Quantity	Calc	Sim	Bench	Gap
1	Load regulation 0.5→2 A (J3)	-0.075 %	—	-0.235 %	-213.8 %
2	Input ripple p-p at C1, 12 V / 2 A (20 MHz BWL)	37.67 mV	97.28 mV	89.84 mV	+138.5 %
3	Load step undershoot, 0.5→1.5 A at 1 A/µs	—	83.21 mV	46.34 mV	-44.3 %
4	Output ripple p-p, 12 V / 2 A (20 MHz BWL)	5.10 mV	4.22 mV	3.91 mV	-23.4 %
5	Switch-node Pk-Pk, 12 V / 2 A	12.7 V	—	14.45 V	+13.8 %
6	Thermal drift, 3 min soak at 2 A	-3.30 mV	—	-3.56 mV	-8.0 %
7	Startup 10–90 % rise time	1.320 ms	1.320 ms	1.422 ms	+7.8 %

#	Quantity	Calc	Sim	Bench	Gap
8	Inductor ripple ΔIL p-p, 12 V / 2 A	0.435 A	0.4575 A	0.4688 A	+7.8 %
9	Efficiency, 12 V / 10 mA	0.8710	—	0.8297	-4.7 %
10	Efficiency, 12 V / 0.2 A	0.8758	0.9763	0.8449	-3.5 %
11	Efficiency, 12 V / 1 A	0.9151	0.9567	0.8935	-2.4 %
12	Duty cycle, 12 V / 2 A	0.2880	0.2911	0.2936	+2.0 %
13	Efficiency, 12 V / 2 A	0.8996	0.9389	0.8831	-1.8 %
14	Average inductor current, 12 V / 2 A	2 A	2 A	2.025 A	+1.3 %
15	Output voltage, 12 V / no load (J3)	3.3 V	3.299 V	3.326 V	+0.8 %
16	Line regulation 8→15 V, 1 A	0.140 %	—	0.139 %	-0.7 %
17	Switching frequency, 12 V / 2 A	2.618 MHz	2.500 MHz	2.636 MHz	+0.7 %

Table 10. Ranked comparison (data/analysis_table.csv).

1. Load regulation 0.5→2 A (J3). The datasheet's typical 0.05 %/A is measured at the IC. At J3 the output copper (2.5 mΩ) is outside the VOS sense point, and as load rises Tj rises (~28.5 °C at 2 A) which pulls Vref down (-35 ppm/°C). Both add to the IC's own regulation error.

2. Input ripple p-p at C1, 12 V / 2 A (20 MHz BWL). Largest gap. The formula only has the capacitive (26.58 mV) and ESR terms. The input current is a pulse train with ~4 ns edges, and the capacitor ESL turns each edge into a voltage spike of ESL·di/dt that survives the 20 MHz filter as a large negative/positive lobe. The capacitive part alone matched the sim to within 10% (28.46 mV).

3. Load step undershoot, 0.5→1.5 A at 1 A/µs. The sim uses a generic voltage-mode Type III loop (~150 kHz crossover) that must wind up an integrator. DCS-Control reacts on the next switching cycle through its direct comparator path, so the real part undershoots less and recovers faster.

4. Output ripple p-p, 12 V / 2 A (20 MHz BWL). The calc includes the ESL step (2.73 mV) that a 20 MHz bandwidth limit mostly removes, so it overpredicts. On the bench, the probe/scope noise floor (measured 1.09 mV p-p with the tip shorted) adds on top, and the higher fsw shrinks the capacitive term. The capacitor-only ripple agreed within a few percent in stage 2 (1.98 mV calc vs 2.03 mV sim).

5. Switch-node Pk-Pk, 12 V / 2 A. Ideal expectation is Vin plus one body-diode drop. The bench shows ringing from the hot-loop inductance (~0.9 nH) resonating with the switch-node capacitance at 531 MHz; the 200 MHz scope attenuates it, so the true overshoot is larger than displayed. The 2 ns sim step cannot resolve it (damped on purpose in stage 2).

6. Thermal drift, 3 min soak at 2 A. Vout drifts down as the junction warms with a ~25 s time constant. Expected size = Vref tempco × ΔTj × Vout. This is why regulation sweeps need a dwell time and a fixed sweep direction.

7. Startup 10–90 % rise time. Rise time follows Css·1.25 V/Iss. The real soft-start capacitor measured high and the SS current source low (both within tolerance), giving tSS = 1.79 ms instead of 1.65 ms.

8. Inductor ripple ΔIL p-p, 12 V / 2 A. Three effects stack: (1) the hand calc used ideal D and 2.5 MHz; (2) the inductor's inductance droops about 6.00% near 2 A (soft saturation), steepening both slopes; (3) the current probe's 6.00 mA rms noise adds roughly 6σ to a Pk-Pk reading. The higher bench fsw works the other way (shorter period, less ripple).

9. Efficiency, 12 V / 10 mA. Deep power-save: loss is dominated by fixed per-pulse and bias terms, which the model underestimates. Below ~10 mA the e-load's programming offset also becomes significant, so output current was taken from the load's readback; a series DMM on the output is the better method for those points.

10. Efficiency, 12 V / 0.2 A. 0.2 A is below the 218 mA power-save threshold. The real part (and the loss model) switch in pulses; the generic sim is forced-CCM and has no gate loss, so it is far too optimistic. The bench adds PSM bias current (60 µA from Vin) and DCM ringing energy lost each pulse.

11. Efficiency, 12 V / 1 A. Same mechanisms as at 2 A; at 1 A the frequency-dependent extras (recovery, core loss) are a bigger share of the smaller total loss, so the percentage gap is similar even though I^2R terms are 4x smaller.

12. Duty cycle, 12 V / 2 A. All three include the resistive drops; the sim and bench also include the dead-time body-diode drop and the extra package/PCB resistance, so D is slightly higher. Scope duty is measured at mid-level on an edge with ringing, which adds ±1 sample (0.2 ns) of uncertainty per edge.

13. Efficiency, 12 V / 2 A. The sim is highest because it has no gate-drive, reverse-recovery or core losses and its soft ~4 ns edges switch with little overlap. The loss model adds gate drive and temperature-scaled Rds(on)/DCR. The bench adds package/PCB copper (12 mΩ), low-side reverse recovery (1.5 nC), inductor core loss (~28 mW at 2.5 MHz) and 2.5 mΩ of output copper before J3.

14. Average inductor current, 12 V / 2 A. The current probe was zeroed/degaussed before the session but still reads 18.00 mA of offset; current probes drift with temperature and after exposure to DC. Physically $I_{L,avg} = I_{out}$.

15. Output voltage, 12 V / no load (J3). Set-point error from real parts, not the design: this board's Vref sits 0.30% high (spec allows ±1.8%), R1 is 0.40% and R2 -0.21% off nominal (1% parts), and 1.5 nA of FB leakage through R1 adds 1.13 mV. The sim uses ideal parts, so it sits exactly on the design value.

16. Line regulation 8→15 V, 1 A. Matches the datasheet typical because nothing on the board adds line-dependent error at 1 A; line regulation is set inside the IC. A small thermal term (more switching loss at 15 V) is within noise.

17. Switching frequency, 12 V / 2 A. The TPS62130A uses a fixed on-time (t_{ON} = 400 ns·Vout/Vin), so $f_{sw} = D/t_{ON}$ rises when losses push D up. The generic sim is fixed-frequency at 2.5 MHz, so it cannot show this; the hand calculation can. The bench adds 12 mΩ of package/PCB resistance, which raises D (and fsw) slightly more.

11. Conclusions

The design met its targets with margin: the inductor peak current was well inside its rating, the junction ran at about 54 °C at full load, and regulation stayed within the datasheet's accuracy window. The largest prediction gaps did not come from arithmetic but from what simple formulas leave out: copper and temperature in load regulation, ESL edges in input ripple, and the control architecture in the load step. The capacitor-only ripple, duty cycle, peak current and line regulation all agreed within a few percent once the model and the measurement were compared at the same physical node and bandwidth.

Lessons: always measure the probe's noise floor before trusting a millivolt ripple reading; compare quantities at the same node and bandwidth; take frequency from the switch node, not a ripple channel; use a dwell time and fixed sweep direction for regulation because the board's temperature follows the load; and treat a simulation result that looks physically impossible as a model problem first.

Next step: a rev C spin adding a second 22 µF output capacitor to reduce output ripple and load-step deviation, a thermal-camera check of the IC and inductor at 2 A, and replacing the approximate datasheet reference points with values digitized from the datasheet figure.